

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	xxxxxxxxxx
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI-powered crop recommendation system using Machine Learning. The system analyzes soil nutrients and environmental conditions to recommend the most suitable crop. It helps farmers make informed decisions and improve agricultural productivity.
Scope	The project collects agricultural data, trains a Machine Learning model, and integrates it into a Flask web application. Users can enter soil and weather parameters to receive real-time crop recommendations through an easy-to-use interface.
Problem Statement	
Description	Farmers often choose crops based on experience rather than scientific analysis, resulting in low yields and financial losses. The lack of intelligent decision-support systems makes accurate crop selection difficult.
Impact	The system improves crop selection accuracy, increases agricultural productivity, reduces crop failure, and supports sustainable farming practices. It enables farmers to make faster and more reliable decisions.

Proposed Solution	
Approach	The application preprocesses agricultural data and uses a trained Machine Learning model to predict the best crop. User inputs are analyzed in real time, and the recommended crop is displayed through a Flask-based web application.
Key Features	1. AI-based crop recommendation. 2. Real-time prediction using soil and weather parameters. 3. User-friendly Flask web interface. 4. Fast and accurate Machine Learning predictions.



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv